

TeamSprintUp

A Simulated Hi-Tech Workplace for Students

THE PROBLEM

- High school students in tech programs gain theoretical knowledge but never experience real hi-tech workplace dynamics.
- They lack exposure to real teamwork, approval pipelines, role ownership, and deadline pressure.
- The gap between the classroom and the industry remains wide.

THE SOLUTION

TeamSprintUp simulates a real hi-tech workplace inside the classroom. Students are divided into teams, assigned professional roles, and complete 3D-printing engineering challenges as real work tasks — with a Monday.com-style management board, a structured approval pipeline, AI-powered hints, and knowledge quizzes.

HOW IT WORKS

- 1 Teacher publishes a challenge**
e.g. "Design a tactile puzzle for visually impaired children"
- 2 Students receive professional roles**
Designer · QA · Editor (Slicer) · Printer
- 3 Each student completes the before-quiz**
Locks in prior knowledge; unlocks the task checklist
- 4 Student executes their role task**
Follows a step-by-step checklist; can request AI hints
- 5 Approval pipeline runs**
QA reviews !' PM approves !' Teacher signs off
- 6 After-quiz + learning gain**
Post-quiz score compared to pre-quiz !' growth metric for teacher

KEY FEATURES

- **Gamification** Bubble-quiz game with 8-bit audio, colored role chips, and step-by-step checklists
- **Approval Pipeline** Dev !' QA !' PM !' Teacher structured sign-off, mirroring real development workflows
- **AI Hints (Claude)** Context-aware hints generated by Anthropic Claude, grounded in the full Hebrew syllabus
- **Teacher Dashboard** Challenge management, manual/auto role assignment, per-student analytics
- **Learning Analytics** Active time, approved tasks, and quiz learning gain per student
- **Role Explanation** "Junior Designer" onboarding popup after every pre-task quiz submission
- **Challenge Gating** Checklist locked until pre-quiz done; after-quiz locked until all steps completed
- **Offline-first POC** Runs fully without a live backend — all demo data built in for hackathon demos

TECHNOLOGY STACK

Frontend	Nuxt 3 · Vue 3 · Tailwind CSS
Backend	NestJS 10 (TypeScript) · REST API
Database	PostgreSQL on Neon Cloud (porsager/postgres)
AI	Anthropic Claude — RAG-grounded hint generation

CURRENT STATUS — DEMO READY

- Full task pipeline with live QA !' PM !' Teacher approval chain
- Teacher dashboard: challenge publishing, role assignment, analytics tab
- Pre/post knowledge quizzes with learning gain calculation
- 8 demo students, 2 teams, 3 real TechSchool 3D-printing challenges seeded
- POC mode runs entirely without backend — built-in demo fixtures
- Role explanation popup with Junior onboarding, bubble quiz game with 8-bit audio

UNIQUE VALUE

TeamSprintUp is not a simulation — it's a real work experience.

Students don't learn about teamwork — they do teamwork. Every role, every approval, and every deadline is real inside the classroom.

VISION

To turn every technology lesson into a real work experience — so every student graduates high school with hands-on hi-tech teamwork skills, a portfolio of completed engineering challenges, and the confidence to join a real team from day one.
